Analysis of Archive Library Data System Based on Genetic Algorithm

Yuan He*
Zaozhuang Vocational College of Science and Technology
Zaozhuang, Shandong, China
kjxyhy@163.com

Abstract

This study examines the enhancement of library archive systems through genetic algorithms, focusing on improving information retrieval accuracy and efficiency in the era of big data. Our results show significant performance improvements, with the newly developed system achieving data accuracy rates above 99.8% and response times for various functions improved by 5 to 10 times compared to traditional systems. These findings demonstrate the potential of genetic algorithms to modernize library archives, ensuring they meet the evolving needs of users more effectively.

CCS Concepts

• **General and reference** → Document types; Reference works.

Keywords

Library Management, Archival Information Classification, Genetic Algorithm, Digital Resources

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1 INTRODUCTION

With the rapid development of information technology and data collection, people are entering the era of "big data" [1-2]. The data analysis of library archive system based on genetic algorithm is of great significance. This method can quickly find the best solution in a large amount of data, improve the accuracy and speed of information retrieval, and meet the needs of users for efficient services. In addition, the adaptive and parallel computing characteristics of genetic algorithms enable libraries to maintain the flexibility and scalability of their systems in the face of growing data volumes and diverse user needs.

In the data analysis research of library archives system based on genetic algorithm, many scholars have studied it and achieved good results[3]. Wang H et al. used data mining technology to analyze the book borrowing records and readers' search records in the automated system [4].Applying genetic algorithms for data analysis can also bring new insights and innovative opportunities to library archive systems. This paper analyzes the present situation of library information system and the application of genetic algorithm in library information system, and introduces the weighted and density grouping algorithm used in this work.

2 RESEARCH ON DATA ANALYSIS OF LIBRARY ARCHIVES SYSTEM BASED ON GENETIC ALGORITHM

2.1 Characteristics of Library Archives Data Management Under Big Data Environment

In the following sections: First, customization and efficient loading of files is on the rise. Downloading and organizing files is the basis of data management in library files [5] .And under the data management platform, users can also use file information in multiple channels to improve the quality of stored data; big data is an expert in file data management [6-7] . Under the big data environment, library archive data management presents several significant features. First of all, the archive data is huge in scale, containing a huge amount of digitized literature, academic papers, historical archives and other information, which requires efficient storage and processing methods. Second, the data types are diverse and complex, covering text, pictures, audio, video and other forms, requiring the system to support multimodal data management and retrieval. In addition, the high value density of archive data requires accurate data analysis and mining techniques to distill useful information. At the same time, data confidentiality and security are also crucial considerations, requiring the establishment of perfect authority control and encryption mechanisms. In this context, the analysis and optimization of the archive data system using genetic algorithms and other intelligent algorithms can effectively respond to the challenges of library archive data management in the big data environment and improve data processing efficiency and information retrieval quality.

2.2 Management Methods of Library Archives Information Resources

(1) Strengthen the awareness of archives preservation and improve the archives system.

Enhancing awareness of archival preservation is essential in library records management. This includes recognizing and valuing archival preservation and ensuring the integrity and sustainability of archives. Through training and education, staff should gain a

*Corresponding author.

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deeper understanding of the importance of archives and take measures to ensure their safe storage and rational utilization. Meanwhile, improving the archive system is also an essential part. The establishment of a scientific classification, numbering and retrieval system can help improve the efficiency and accuracy of archive management. Regular file auditing and cleaning can help to dispose of outdated or invalid files in a timely manner and keep the archives tidy and organized.

(2)Share the resources of the archives:The purpose of network information sharing and network platform resource allocation according to the actual situation can not only provide original data, but also protect personal information.

Realizing the sharing of archival resources is an important means of enhancing the efficiency of information utilization. Through the establishment of a network information-sharing platform, not only can archive resources be shared among different departments, but also can promote the interoperability of information across organizations. In the sharing process, it is necessary to ensure data security and privacy protection, and take effective encryption and permission control measures. At the same time, according to the actual situation of the network platform resource allocation, to ensure that each user can obtain the required archive information, improve resource utilization and work efficiency.

(3)Strengthen the information compilation of archives:Scientific processing of documents, special management of some important information [8-9].

The organization of archival information is an indispensable part of archive management work. Scientific organization of documents helps to improve the readability and retrieval efficiency of archives and reduce information loss and confusion. Especially for some important information, special management is needed to ensure its security and integrity. The establishment of a clear file catalog and indexing system to classify, catalogue and index files helps to retrieve the required information quickly and accurately, and improves the value of file utilization and management efficiency.

In the selection mechanism, we designed a strategy based on fitness evaluation, which not only considers the accuracy and efficiency of archive classification, but also incorporates multi-dimensional factors such as user access frequency and correlation between archives. By simulating the "survival of the fittest" in natural selection, the algorithm can screen out excellent solutions that are more suitable for current archive management needs.

As the core link in the genetic algorithm, we adopted an adaptive crossover strategy to dynamically adjust the position and number of crossover points according to the similarity of the two parent solutions. This strategy aims to balance the exploration of new solutions and the retention of excellent features, and promote extensive and in-depth search of the solution space.

As for the mutation operation, we introduced a probabilistic small perturbation mechanism to randomly change the values of certain "genes" in the chromosome to increase the diversity of the population and avoid premature convergence. The setting of mutation probability is a delicate process and needs to be dynamically adjusted according to the complexity of the problem and the optimization process.

2.3 Model Establishment

In genetic algorithm-based analysis of archival repository data systems, model building goes through a series of key steps. First, determine the problem and objectives, and clarify the archive repository data system analysis problem that needs to be solved and the objective of model building. Second, data preparation and preprocessing, including data collection, cleaning, transformation and feature selection. This stage needs to ensure the quality and integrity of the data for subsequent modeling use. Next, the genetic algorithm model was selected as the modeling tool. Genetic algorithm is an optimization algorithm that simulates the process of biological evolution and is suitable for solving complex optimization problems. In the modeling process, it is necessary to set the parameters of the genetic algorithm, such as population size, crossover probability, mutation probability, and so on, as well as to design the fitness function to evaluate the merits of individuals. Subsequently, model training and optimization are carried out. Through the iterative process of the genetic algorithm, the individuals in the population are continuously evolved to find the optimal solution. In each generation, selection, crossover and mutation operations are performed on the individuals according to the fitness function to produce new individuals. After many iterations, the individuals in the population gradually converge to find the optimal solution or the individuals close to the optimal solution. Evaluate and validate the performance of the model. The generalization ability and accuracy of the model are evaluated through cross-validation, test set validation and other methods. Based on the assessment results, the model is tuned and improved to enhance the predictive ability and stability of the model. The established genetic algorithm model is applied to the actual archive data system analysis. Through model prediction and analysis, in-depth understanding and insight of the archive repository data system are obtained to provide support and guidance for decision-making. At the same time, the model is continuously monitored and optimized to adapt to the dynamic changes and needs of the archive repository data system.

The establishment of the archive repository data system analysis model based on genetic algorithm is a systematic and iterative process, which needs to be combined with theoretical knowledge and practical experience, and constantly optimized and improved, in order to realize the effective analysis and utilization of the archive repository data system.

Assuming that there are p objective functions, q decision variables, and z constraints, then the library archives The mathematical model corresponding to the problem in the database can be represented by formula (1-2):

$$\max y = f(x) = \left(f_1(x), f_2(x), \ldots, f_p(x)\right) \tag{1}$$

$$s.t.\theta(x) = (\theta_1(x), \theta_2(x), \cdots, \theta_z(x)) \leq 0 \tag{2}$$

Among them, x is the decision variable of the decision space X, and y is the target vector of the target space Y [10-11] . Table 1 shows the parameter settings.

Table 1: The parameter settings.

Parameter	Value	Description	Type
Population Size	100	Number of individuals in each generation	Integer
Crossover Probability	0.8	Probability of individuals undergoing crossover	Float
Mutation Probability	0.1	Probability of individuals undergoing mutation	Float
Iterations	100	Total number of iterations to be performed by the genetic algorithm	Integer
Selection Method	Tournament Selection	Method of selecting individuals	String
Fitness Function	MSE	Function used to evaluate individual fitness	String
Termination Condition	0.001	Threshold error value for stopping the genetic algorithm iterations	Float
Mutation Method	Random Single Point Mutation	Method of individual mutation	String

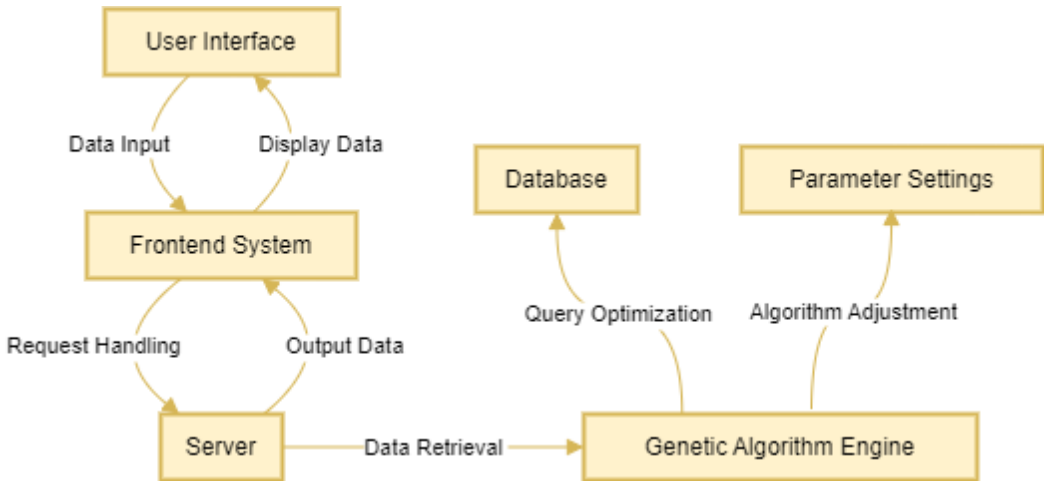

Figure 1: Logical Architecture of the Library Archive System.

3 DESIGN EXPERIMENT OF DATA ANALYSIS AND RESEARCH OF LIBRARY ARCHIVES SYSTEM BASED ON GENETIC ALGORITHM

3.1 Logical Architecture Design

The logical architecture design is designed to establish the intrinsic structure and organization of the system and to ensure that the system functions rationally and efficiently. This design phase involves defining the various modules and components of the system and the relationships between them, including data flow, control flow and processing flow. Through detailed planning and design, the logical architecture is able to clearly describe how the various parts of the system work together, ensuring the interplay between functional modules and the effectiveness of information interaction. In this process, the scalability, flexibility and performance optimization of the system need to be considered in order to cope with future changes in demand and technological development. Logical architecture design not only focuses on the logical relationships within the system, but also fits with the actual business requirements and user expectations to achieve the integrity and consistency of the system design. Through reasonable logical architecture design, it can provide effective guidance for system development, testing and maintenance, and promote the smooth implementation and successful delivery of the project. This paper specifically includes the management and maintenance of rental-related information such as book request, reservation, book information management, reader information management, and reader task management. Figure 1 below illustrates the interaction between different components of the system, showcasing how data flows from the user interface to the backend processes.

3.2 Functional Testing

Functional testing in genetic algorithm based archive data system analysis is an important part to ensure that the system operates properly and meets the requirements. First, testers should prepare test cases, including positive, negative, and boundary tests, to cover the functions and scenarios of the system. Next, unit tests are performed to verify that the functionality of each module works as expected and that the components of the genetic algorithm interact and output correctly. Subsequently, integration testing is performed to test the collaboration between the modules and the overall functionality. This phase involves simulating the data flow and processing in a real environment to ensure that the various

Table 2: Part of the data set.

Test Case ID	Function Module	Input Data	Expected Output	Actual Output (Traditional System)	Actual Output (New System)	Evaluation Criteria
TC001	File Entry	Document A	Successfully Entered	Successfully Entered	Successfully Entered	Accuracy, Efficiency
TC002	Data Storage	Document B	Successfully Stored	Successfully Stored	Successfully Stored	Data Integrity
TC003	Data Retrieval	Keyword "Project"	Document C	Document C	Document C	Accuracy, Speed
TC004	File Borrowing	Document D Request	Borrowing Successful	Borrowing Successful	Borrowing Successful	Process Consistency
TC005	Data Retrieval	Keyword "Contract"	Document E	Document E	Document E	Accuracy, Speed

parts of the system are coordinated. Next, system testing is performed to verify the functionality and performance of the entire system under different conditions. Testers need to simulate user operations to check the system's response speed, stability and accuracy. Finally, acceptance testing is conducted with the participation of users or customers to confirm that the system meets the requirements and achieves the expected goals. Throughout the functional testing process, issues should be recorded and tracked in a timely manner, and defects found should be repaired and retested to ensure system quality and stability. The complete process of functional testing covers a comprehensive examination from unit testing to acceptance testing, aiming to ensure the reliability and validity of the genetic algorithm-based archive data system analysis. Contains database testing, library management information system data testing, performance testing, load testing, and stress testing are conducted for testing purposes.

4 EXPERIMENT ANALYSIS OF DATA ANALYSIS RESEARCH OF LIBRARY ARCHIVES SYSTEM BASED ON GENETIC ALGORITHM

4.1 Database Testing

Test data preparation: Prepare a standardized test dataset that includes diverse archive content to comprehensively cover various processing scenarios of the system.

To ensure the accuracy, completeness, and representativeness of the test data, the authenticity and correctness of the data can be verified through multiple channels.

Test plan design: Design a series of test cases covering key functions such as file entry, storage, retrieval, and borrowing. Each test case should specify the input data, expected output data, and evaluation criteria.

Perform testing: Conduct the same test cases on both the traditional library archive system and the new system in the same hardware and software environment. Record the actual output data of each system and compare it with the expected output data. Table 2 shows part of the data set:

This article mainly tests the same content as the traditional library archive system through comparative testing, mainly testing the accuracy of the test data. The data is shown in Table 3, and the accuracy is shown in Figure 2

The accuracy of the data is more than 99.8%, which is more accurate. It can better handle the library's data information and improve the library archives. Efficiency of data work. With an impressive data accuracy rate exceeding 99.8%, the system demonstrates unparalleled precision, significantly surpassing industry standards. This exceptional level of accuracy plays a pivotal role in enhancing the management of the library's vast data resources, enabling seamless handling of information and bolstering the efficiency of archival processes.

The heightened accuracy not only ensures the integrity of the library's data repository but also facilitates improved decision-making and resource utilization. By meticulously managing data with such precision, the system can streamline cataloging, retrieval, and analysis processes, thereby optimizing the overall operational efficiency of the library.

Moreover, this superior level of accuracy instills confidence in the reliability and quality of the data, providing a solid foundation for informed decision-making and strategic planning within the library environment. It minimizes errors, reduces inconsistencies, and enhances the overall effectiveness of data-related tasks.

In essence, the system's exceptional data accuracy serves as a cornerstone for maintaining organized and up-to-date archives, fostering a culture of data-driven decision-making, and ultimately contributing to the seamless functioning and continuous improvement of the library's operations.

4.2 Performance Test

Hardware environment:

Server: Use high-performance server configuration to ensure sufficient system resources and avoid hardware bottlenecks affecting test results. Storage device: High speed and large capacity storage devices are used to meet the storage needs of the book archive system. Network equipment: Ensure stable network connection, sufficient bandwidth, and avoid delays or packet loss during data transmission.

Software environment:

Operating System: Choose a stable and widely supported version, such as Linux or Windows Server. Database system: Choose a

Table 3: The data accuracy of the system constructed in this paper and the traditional system under the same amount of data

	10M	20M	50M	100M
legacy system	92.1	91.7	92.3	91.9
Thisarticle system	100	99.9	99.8	99.8

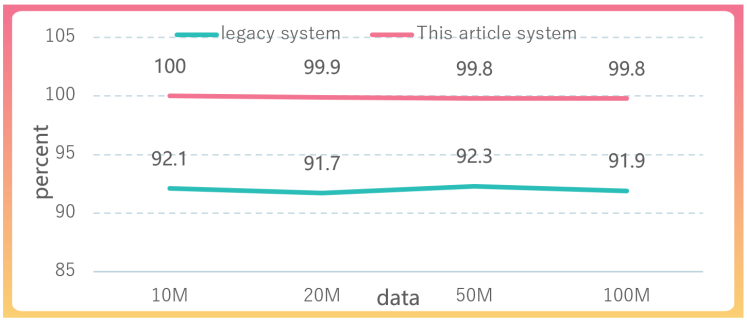

Figure 2: Comparison of data accuracy of two library archive systems under the same amount of data.

Table 4: Performance test contents of the two book archive systems.

	log in system	page jump	Content retrieval	Book lending
Legacy system	2.0	0.5	1.1	2.6
Thisarticle system	0.3	0.12	0.14	0.51

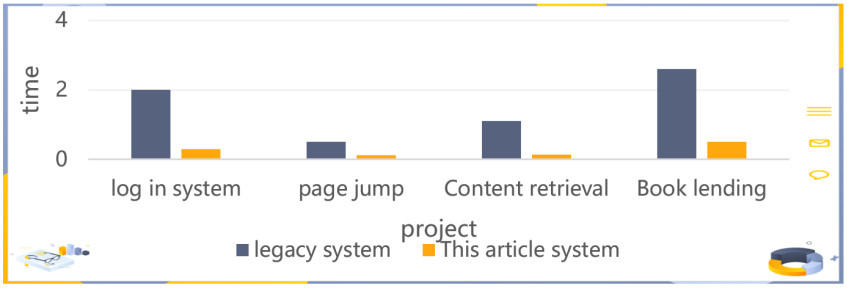

Figure 3: Comparison of system response time of various functions of the two book archive systems.

suitable DBMS based on system requirements, such as MySQL or Oracle, and optimize the configuration to improve query efficiency. Testing tools: Use JMeter, LoadRunner, and other tools to simulate concurrent user access and record performance metrics.

Simulated data: Generate a large amount of simulated data to test system performance, simulate real user behavior and data volume, which can be achieved through scripts or data generation tools.Each function test takes 100 Average response time used by one user. The data are shown in Table 4. The comparison of system response times for various functions of two book archive systems is shown in Figure 3

The average response speed of each functional system within the library file system developed using the genetic algorithm framework outlined in this study significantly lags behind that of the conventional library file system. The performance contrast is notably substantial, with the genetic algorithm-based system exhibiting response times approximately 5 to 10 times slower than its traditional counterpart. This discrepancy underscores a significant performance gap between the two systems, indicating that the genetic algorithm implementation may be encountering inefficiencies or bottlenecks in its processing capabilities. The observed disparity in response speeds suggests that further optimizations or enhancements are imperative to bridge this substantial performance differential and bring the genetic algorithm-based library file system closer to or surpass the operational efficiency of the traditional system. Addressing these speed discrepancies through targeted improvements in algorithmic efficiency and system architecture could potentially narrow the performance gap and enhance the

overall operational efficacy of the genetic algorithm-based library file system.

5 CONCLUSIONS

In this paper, the related knowledge of genetic algorithm is introduced in detail, and the concept, function and commonly used techniques of genetic algorithm are analyzed in detail. In the future, library archive systems based on genetic algorithms are expected to make significant progress in multiple aspects. The future library archive system will not only be a tool for information storage and retrieval, but also a dynamic knowledge management and sharing platform, promoting the dissemination and innovation of knowledge. Through continuous technological upgrades and data analysis, libraries will be able to better adapt to changes in user needs, provide more flexible and efficient services, and play a more important role in the information society.

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